

ASCII MASTER FLOW – PANEL 1/12: Nanoscale taxonomy and property gateway

1-100 nm -> functional size-dependent behaviour

3 NANOSCALE DIMENSIONS -> nanoparticle

2 NANOSCALE DIMENSIONS -> nanotube / nanofibre

1 NANOSCALE DIMENSION -> nanosheet / nanoplate

GATEWAY -> surface-area effect + possible quantum effect

ASCII MASTER FLOW – PANEL 2/12: Surface-area and quantum twin engine

SMALLER SIZE -> larger relative surface -> adsorption / catalysis / reactivity

SUFFICIENT CONFINEMENT -> altered states / band gap -> optical or electronic shift

MATERIAL + SIZE + SURFACE CHEMISTRY -> actual behaviour

TRAP -> nano is neither an automatic benefit nor safety verdict

ASCII MASTER FLOW – PANEL 3/12: Fabrication route comparison

TOP-DOWN -> bulk -> lithography / etching / milling -> patterned structure

BOTTOM-UP -> atoms / molecules -> self-assembly / sol-gel / CVD -> formed structure

TOP-DOWN LIMIT -> waste / defects / feature constraint

BOTTOM-UP LIMIT -> aggregation / uniformity / purification

COMMON TEST -> reproducibility and scale-up

ASCII MASTER FLOW – PANEL 4/12: Nanomaterial geometry atlas

PARTICLE -> three nanoscale external dimensions

CNT -> cylindrical carbon nanostructure

GRAPHENE -> single-atom-thick hexagonal carbon sheet

QUANTUM DOT -> size-tunable semiconductor nanocrystal

NANOCOMPOSITE -> nanoscale phase modifies bulk matrix

ASCII MASTER FLOW – PANEL 5/12: Characterization decision tree

SURFACE MORPHOLOGY? -> SEM

INTERNAL STRUCTURE / NANOSCALE DIMENSION? -> TEM

PROBE-TRACED TOPOGRAPHY? -> AFM

CRYSTALLINE PHASE? -> XRD

DISPERSION SIZE / STABILITY? -> DLS + zeta potential

ASCII MASTER FLOW – PANEL 6/12: Nanomedicine barrier rail

CARGO -> encapsulation / protection

CARRIER -> circulation and release profile

TARGETING -> passive accumulation or ligand interaction

BIOLOGICAL BARRIERS -> tissue -> cell -> intracellular site

STATUS -> delivery evidence != approval / superiority / safety

ASCII MASTER FLOW – PANEL 7/12: Agriculture value chain

NANO-INPUT / SENSOR / SEED COATING -> farm decision or delivery

FIELD EVIDENCE -> dose, crop, soil, weather and formulation

COST + EXTENSION + CREDIT -> adoption boundary

INPUT LOSS / SPOILAGE / STRESS DETECTION -> possible income channel

VERDICT -> socio-economic uplift must be demonstrated

ASCII MASTER FLOW – PANEL 8/12: Water and environment treatment loop

CONTAMINANT -> detection / adsorption / membrane / photocatalysis

SYSTEM TEST -> selectivity + fouling + energy

RECOVERY -> regeneration or spent-material capture

LIFECYCLE CHECK -> secondary nanoparticle release

BOUNDARY -> lab removal != field deployment

ASCII MASTER FLOW – PANEL 9/12: Electronics energy materials matrix

GRAPHENE / CNT -> conductive and structural research

QUANTUM DOT -> optical / display / sensing research

NANOCOMPOSITE -> coatings and lightweight-strength modification

NANOSTRUCTURED ELECTRODE -> storage and conversion interfaces

BOUNDARY -> property != certified manufacturable product

ASCII MASTER FLOW – PANEL 10/12: Indian nano institution map

DST NPNST / EARLIER NANO MISSION -> public R&D architecture

INST MOHALI -> DST autonomous nanoscience institution

CSIR LABS -> cross-sector translational capability route

MeitY-LINKED NANO-ELECTRONICS -> device / semiconductor bridge

FIREWALL -> page / call / event != programme outcome

MUST REMEMBER: Nanotechnology works through size-dependent surface, quantum, optical,...

ASCII MASTER FLOW – PANEL 11/12: Toxicity and ecotoxicity exposure web

MATERIAL -> size + shape + coating + solubility + aggregation

EXPOSURE -> dose + route + duration + susceptible system

ENVIRONMENT -> transformation + mobility + persistence

ECOLOGY -> organism effects + food-web transfer + bioaccumulation

RULE -> material- and use-specific evidence

CLOSE DISTINCTION: Nanoscale is not automatically novel or safer, nanoparticle is not...

ASCII MASTER FLOW – PANEL 12/12: Regulatory and evidence status ladder

SYNTHESIS -> CHARACTERIZATION -> LAB FUNCTION

ANIMAL / GREENHOUSE -> FIELD / CLINICAL STUDY

SECTORAL REVIEW -> STANDARD / NOTIFICATION / APPROVAL

MANUFACTURE -> MARKETING -> ADOPTION

MONITORED OUTCOME -> only dated evidence supports this rung

EVIDENCE LIMIT: MECHANISM / STATUS / CAUSATION: Give particle/material identity, size...